

Submission Summary

Conference Name7th International Conference on Data Science and Applications

Paper ID2058

Paper TitleCMB Power Spectrum Analysis using Planck 2018 Data for Early Universe Cosmology

Abstract

The Cosmic Microwave Background (CMB) encodes primordial density fluctuations from approximately 380,000 years after the Big Bang and provides stringent constraints on Λ CDM cosmological parameters. This work presents a computational pipeline for analyzing Planck 2018 SMICA temperature maps and reconstructing the angular power spectrum of CMB anisotropies. The temperature field is decomposed using spherical harmonics, and the power spectrum is computed via the HEALPix anafast routine, expressed as $D_\ell = \ell(\ell+1)C_\ell / 2\pi$. Galactic foreground contamination is mitigated using a binary mask, retaining approximately 70–75% of the sky.

The reconstructed spectrum over multipole range $\ell \leq 2000$ exhibits characteristic acoustic peaks at $\ell \approx 220, 540$, and 815 , consistent with photon–baryon oscillations in the early universe. Theoretical spectra generated using the CAMB Boltzmann solver with Planck 2018 best-fit parameters ($H_0 = 67.4$ km/s/Mpc, $\Omega_b h^2 = 0.0224$, $\Omega_c h^2 = 0.120$, $n_s = 0.965$) show strong agreement with observational results across angular scales. Additionally, cosmological parameters were estimated using a simplified Markov Chain Monte Carlo (MCMC) approach over multipoles $\ell \leq 800$, yielding values consistent with Planck constraints.

These results demonstrate consistency with the Λ CDM cosmological model and show that meaningful cosmological analysis can be performed using publicly available datasets and accessible computational tools.

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Submission Files

CMB_Power_Spectrum_Analysis_Planck2018.pdf (606.6 Kb, 2026-05-15 00:22:54 GMT+5:30)

Submission Questions Response

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The author confirms sole responsibility for the following: study conception and design, data collection, analysis and interpretation of results, and manuscript preparation.

Study conception and design: Romit Ghosh;

Data collection: Romit Ghosh;

Analysis and interpretation of results: Romit Ghosh, Prabhat Kumar;

Manuscript preparation: Romit Ghosh;

Technical assistance: Prabhat Kumar.

All authors reviewed the results and approved the final version of the manuscript.

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